MATHEMATICS / INDEPENDENT STUDY

MM7030

Category Theory

Written lectures · eight teaching units

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

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1 Unit 1 · Categories and duality

1.1 Lecture 1 · Objects, arrows, and composition

Learning goals and prerequisites
Verify category axioms and recognize algebraic examples. Recall sets, functions, groups, and homomorphisms.

Recall and approach. Recall that a function includes a domain and codomain, and a monoid needs associativity and an identity but not inverses.

Many mathematical constructions are determined less by what their elements look like than by the maps they admit. A product, for example, receives pairs of maps. Category theory makes these mapping patterns explicit. The first task is not abstraction for its own sake: it is to name exactly which maps preserve the structure under discussion.

Definition 1 · Category
A category $\mathcal{C}$ has objects, a set $\mathcal{C}(A, B)$ of morphisms for each pair of objects, identities $1_A \in \mathcal{C}(A, A)$, and associative composition $\mathcal{C}(B, C) \times \mathcal{C}(A, B) \to \mathcal{C}(A, C)$. The identity laws are $1_B f = f = f 1_A$. We write gf for $g \circ f$.

We work with locally small categories, meaning each hom collection is a set. A category is small when its objects and arrows together form sets. The category of all sets is not small under the usual set-theoretic foundations. All diagrams used for limits here have a small indexing category. This distinction prevents forming an illegitimate "set of all sets."

Example 1 · Four families
In Set, objects are sets and arrows are functions. In Grp, they are groups and homomorphisms. A poset P gives a category with one arrow $a \to b$ exactly when $a \leq b$. A monoid gives a category with one object whose endomorphisms are the monoid elements and whose composition is multiplication.

For a poset, composition is supplied by transitivity, identity by reflexivity, and associativity is automatic because there is at most one arrow of any given type. Conversely, a one-object category is a monoid because its composition and identity satisfy exactly the monoid axioms. A group is the special case in which every arrow is invertible.

Definition 2 · Isomorphism
A morphism $f : A \to B$ is an isomorphism if some $g : B \to A$ satisfies $gf = 1_A$ and $fg = 1_B$.

Proof strategy. Check the types first, then compare two proposed inverses using associativity.

Theorem 1 · Uniqueness of an inverse
An isomorphism has a unique inverse.

Proof. If g, h both invert f, then $g = g(fh) = (gf)h = h$. Every composition in this equation is defined because the domains and codomains match. □

Watch for this. Objects being isomorphic does not mean they are equal. Two different two-element sets are isomorphic in Set. Category theory often specifies an object only up to isomorphism, and a later universal property will make the relevant isomorphism unique.

Example 2 · An arrow can carry more information than a formula
The expression $x \mapsto x^2$ can define a function $\mathbb{R} \to \mathbb{R}$ or a function $\mathbb{R} \to \mathbb{R}_{\geq 0}$. The formulas agree, but the arrows have different codomains and therefore different composability and surjectivity properties. To check a proposed category of structured objects, first specify each hom set, then verify that the proposed composite remains inside the required hom set. For groups, this closure follows from $(gf)(ab) = g(f(a)f(b)) = gf(a)gf(b)$. Associativity is inherited from functions. Restricting the arrows without checking this closure can destroy the category.

Proof-writing habit. Above every composite, temporarily write its source and target. The identity in $gf = 1_A$ is forced by those types; it cannot be replaced by an unindexed symbol until the typing is secure.

Checkpoint 1.1. When are two objects isomorphic in the category associated with a poset?

Takeaways. Keep a type ledger for arrows. Many apparent algebraic identities fail simply because a composite is not defined.
Continue with study guide U1 and exercise sheet U1.

1.2 Lecture 2 · Duality, monomorphisms, and epimorphisms

Learning goals and prerequisites
Reverse statements correctly, apply cancellation definitions, and recognize where elementwise intuitions fail.

Recall and approach. Recall injective and surjective functions and the use of a singleton to select an element.

Definition 3 · Opposite category
The opposite category $\mathcal{C}^{\mathrm{op}}$ has the same objects and $\mathcal{C}^{\mathrm{op}}(A,B) = \mathcal{C}(B,A)$. Composition reverses the order inherited from $\mathcal{C}$. Reversing twice returns the original category.

A theorem proved solely from category axioms has a dual theorem obtained by reversing all arrows. Duality is not replacing every function by an inverse: most arrows have no inverse. It changes the category in which the statement is read.

Definition 4 · Mono and epi
A morphism $m : A \to B$ is monic if $mu = mv$ implies $u = v$ for all parallel $u, v : X \to A$. A morphism $e : A \to B$ is epic if $ue = ve$ implies $u = v$ for all parallel $u, v : B \to X$.

Proof strategy. To prove cancellation, start with an arbitrary pair of parallel arrows. To refute it, construct one pair that becomes equal after composition.

Theorem 2 · Cancellation in sets
Monomorphisms in Set are precisely injections; epimorphisms are precisely surjections.

Proof. An injective map cancels on the left pointwise. If m is not injective, two distinct elements with the same image define different maps from a singleton which m identifies, so it is not monic. A surjection cancels on the right because every target element has a preimage. If e is not onto, choose a missing target element and define two maps to $\{0,1\}$ which agree everywhere except there; they agree after precomposition with e but are different. □

Theorem 3 · Split arrows
If $rs = 1_A$ for $s : A \to B$ and $r : B \to A$, then s is monic and r is epic.

Proof. If $su = sv$, compose with r to obtain $u = v$. If $ur = vr$, compose with s to obtain $u = v$. □

Example 3 · Mono and epi need not mean isomorphism
In the category of commutative unital rings, the inclusion $\mathbb{Z} \to \mathbb{Q}$ is monic because it is injective. It is epic: two maps $\mathbb{Q} \to R$ agreeing on integers must agree on $\frac{a}{b}$, whose image is forced to be the image of a times the inverse of the image of b. But the inclusion is not surjective, hence not an isomorphism.

This example explains why the definitions use cancellation rather than the words "injective" and "surjective." A category may have objects that are not sets equipped with structure, or its arrows may detect only some distinctions between elements.

Example 4 · Cancellation without enough arrows to detect elements
In the poset with $0 < 1$, its nonidentity arrow is both mono and epi: every pair of parallel arrows already agrees. It has no inverse, since there is no arrow $1 \to 0$. This gives a very small example where categorical cancellation says less than invertibility. By contrast, in sets, maps from a singleton detect unequal elements and maps to a two-element set detect missing elements. The surrounding category controls which tests exist.

Duality drill. A left inverse proves monicity by composing on the left. Reverse every arrow to obtain: a right inverse proves epicity by composing on the right. The proof is transported, not repeated element by element. Do not dualize "injective function" into a vague opposite operation; dualize the precise cancellation equation.

Checkpoint 1.2. Is every arrow in a poset category both monic and epic?

Takeaways. Translate a definition into a diagram and then dualize the diagram. Use set intuition to discover examples, not as a replacement for the categorical definition.
Continue with study guide U1 and exercise sheet U1.

2 Unit 2 · Functors and equivalences

2.1 Lecture 3 · Functors as structure-preserving translations

Learning goals and prerequisites
Define functors on both objects and arrows, verify their laws, and distinguish covariant from contravariant constructions.

Recall and approach. Recall that precomposition changes the source of a map and postcomposition changes its target.

Definition 5 · Functor
A functor $F : \mathcal{C} \to \mathcal{D}$ assigns an object FA to each A and an arrow $Ff : FA \to FB$ to each $f : A \to B$, preserving identities and composition: $F(1_A) = 1_{FA}$ and $F(gf) = F(g)F(f)$. A contravariant functor on $\mathcal{C}$ is a covariant functor from $\mathcal{C}^{\mathrm{op}}$.

The forgetful functor $U : \mathrm{Grp} \to \mathrm{Set}$ keeps the underlying set and function. The free vector-space functor sends a set X to finite formal linear combinations of its elements; a function $f : X \to Y$ sends each basis vector $[x]$ to $[f(x)]$ and extends linearly. Preservation of composition follows by checking on basis vectors, which determine a linear map uniquely.

Example 5 · Power sets in two directions
Direct image gives a covariant functor $\mathcal{P} : \mathrm{Set} \to \mathrm{Set}$, since $g[f[S]] = (gf)[S]$. Inverse image gives a contravariant functor, since $(gf)^{-1}[T] = f^{-1}[g^{-1}[T]]$. The same object assignment supports two different arrow assignments.

Proof strategy. Apply the functor to the inverse equations. For examples, check identities and composites on generators or individual elements.

Theorem 4 · Functors preserve isomorphisms
If f has inverse g, then Ff has inverse Fg.

Proof. The functor laws give $FgFf = F(gf) = F(1_A) = 1_{FA}$ and likewise in the other order. □

For a fixed object A, the covariant hom functor $\mathcal{C}(A, -)$ sends X to its maps from A and sends $f : X \to Y$ to postcomposition $h \mapsto fh$. The contravariant hom functor $\mathcal{C}(-, A)$ sends $f : X \to Y$ to precomposition $k \mapsto kf$. Associativity proves the functor laws. These constructions will become the main objects of the Yoneda lemma.

Watch for this. A functor need not preserve monomorphisms, epimorphisms, or limits merely because it preserves composition. Those claims require additional hypotheses. Preservation of isomorphisms is special because an inverse is expressed entirely by composition and identity equations.

Example 6 · Checking a dual map with a type ledger
Given $f : V \to W$, a functional $\varphi : W \to K$ can be precomposed to give $\varphi f : V \to K$. Thus $f^* : W^* \to V^*$. For $g : W \to X$ and $\psi : X \to K$, $(gf)^*(\psi) = \psi gf = f^*(g^*(\psi))$, so $(gf)^* = f^*g^*$. The reversed order is forced by the domains, not an arbitrary convention. The identity calculation is $\varphi 1_V = \varphi$.

Counterexample to a careless recipe. Sending every set to itself and every map to an identity cannot define an endofunctor on sets: a map between two different objects would need an arrow between those

two images, whereas an identity has only one object as both source and target. Arrow typing precedes the functor laws.

Checkpoint 2.1. What direction does the dual-vector-space construction send a linear map $f : V \to W$?

Takeaways. An object recipe alone is not a functor. Define the action on arrows, check types, then verify the two laws.
Continue with study guide U2 and exercise sheet U2.

2.2 Lecture 4 · Fullness, faithfulness, and equivalence

Learning goals and prerequisites
Distinguish an equivalence from equality of categories and prove the standard equivalence criterion.

Recall and approach. Recall bijections on hom sets and the distinction between an object being equal and being isomorphic to another.

Definition 6 · Full, faithful, essentially surjective
A functor is faithful if each map $\mathcal{C}(A, B) \to \mathcal{D}(FA, FB)$ is injective, full if each is surjective, and essentially surjective if every object of $\mathcal{D}$ is isomorphic to some FA.

The forgetful functor from groups to sets is faithful: a group homomorphism is determined by its underlying function. It is not full, because arbitrary functions need not preserve multiplication. A full subcategory retains all arrows between its chosen objects; a subcategory need not be full.

Definition 7 · Equivalence
An equivalence consists of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong 1_{\mathcal{C}}$ and $FG \cong 1_{\mathcal{D}}$. Naturality, developed next, means these isomorphisms respect every arrow.

Proof strategy. Choose representatives and comparison isomorphisms, lift arrows by fullness, and prove equations by faithfulness.

Theorem 5 · Equivalence criterion
A functor is an equivalence if and only if it is full, faithful, and essentially surjective, with a choice of representatives when required.

Proof. Suppose F has these three properties. For each D, choose GD and an isomorphism $\varepsilon_D : FGD \to D$. Fullness and faithfulness define a unique Gf by $F(Gf) = \varepsilon_E^{-1} f \varepsilon_D$ for $f : D \to E$. This formula and faithfulness prove the functor laws. The chosen ε is natural by construction. For C, fullness gives a unique $\eta_C : C \to GFC$ whose image is ε_{FC}^{-1}. Full faithfulness reflects inverses, so η_C is an isomorphism; the same formula proves its naturality. Conversely, a quasi-inverse and natural isomorphisms give inverse correspondences on hom sets by conjugating with their components, so F is full and faithful. The isomorphism $FGD \cong D$ gives essential surjectivity. □

Example 7 · A skeleton of finite-dimensional vector spaces
The category of finite-dimensional K-vector spaces is equivalent to the category with objects nonnegative integers and arrows $n \to m$ the m-by-n matrices. Send n to K^n and a matrix to its linear map. This functor is full and faithful; every finite-dimensional space has a basis and is isomorphic to some K^n. No basis is selected canonically for every space.

Example 8 · Why choosing bases is enough but not canonical
To give a quasi-inverse to the coordinate-space inclusion, select a basis for each finite-dimensional vector space. A linear map is then sent to its matrix in the selected source and target bases. The formula for composition is matrix multiplication, and a change of selected bases conjugates the coordinate description by comparison isomorphisms. This does not make all vector spaces literally equal to coordinate spaces; it makes their categories equivalent.

The converse criterion in detail. If GF and FG are naturally isomorphic to identities, then both F and G are faithful by applying the other functor and conjugating with those isomorphisms. Given $h : FA \to FB$, choose $\eta : 1 \Rightarrow GF$ and put $k = \eta_B^{-1} G(h) \eta_A$. Naturality of η gives $GF(k) = G(h)$; faithfulness of G yields $F(k) = h$. Thus F is full as well.

Checkpoint 2.2. Why is essential surjectivity weaker than surjectivity on objects?

Takeaways. Equivalence preserves the mathematics visible through arrows while allowing multiple isomorphic presentations of the same structure.
Continue with study guide U2 and exercise sheet U2.

3 Unit 3 · Natural transformations

Bridge into this unit. A componentwise formula becomes a natural transformation only after checking it against every allowed arrow. Begin by typing the two routes around a square. The resulting equation expresses uniformity, not an additional formula to memorize.

3.1 Lecture 5 · Naturality is a compatibility equation

Learning goals and prerequisites
Write and check naturality squares, distinguish arbitrary component maps from natural transformations, and recognize natural isomorphisms.

Recall and approach. Recall functoriality and equality of functions established by evaluation on every input.

Two functors can translate the same category in different ways. A natural transformation compares them coherently: it must make the comparison commute with every morphism, not merely supply one map for each object.

Definition 8 · Natural transformation
For $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\eta : F \Rightarrow G$ assigns $\eta_A : FA \to GA$ to each object, such that $G(f)\eta_A = \eta_B F(f)$ for every $f : A \to B$.

$$\begin{array}{ccc} FA & \to \eta_A & GA \\ \downarrow Ff & & \downarrow Gf \\ FB & \to \eta_B & GB \end{array}$$

The square is an efficient memory aid: follow the two routes from FA to GB. This also detects a misplaced functor or component index before any calculation begins.

Example 9 · Evaluation into the double dual
For a vector space V, define $\eta_V : V \to V^{**}$ by $\eta_{V(v)}(\varphi) = \varphi(v)$. For $f : V \to W$, the double dual sends Φ to the functional $\psi \mapsto \Phi(\psi f)$. Thus $f^{**}\big(\eta_{V(v)}\big)(\psi) = \psi(f(v)) = \eta_{W(f(v))}(\psi)$. This is natural. For finite-dimensional spaces it is an isomorphism.

The finite-dimensional assertion follows because a basis and its dual basis show injectivity and equal dimension. Naturality of the map itself requires no basis. This is why the double-dual identification is canonical while an isomorphism between V and V^* constructed from a chosen basis generally is not.

Proof strategy. Compare the two routes through the square at a general arrow; invert components only after writing that equation.

Theorem 6 · Pointwise criterion for natural isomorphisms
A natural transformation is invertible if and only if every component is an isomorphism.

Proof. An inverse transformation supplies componentwise inverses. Conversely, if every η_A is invertible, rearrange $Gf\eta_A = \eta_B Ff$ to get $Ff\eta_A^{-1} = \eta_B^{-1}Gf$. This is exactly the naturality of the family of inverse components. Their compositions with η are identity transformations. □

Watch for this. "There exists an isomorphism for each object" does not establish naturality. One must either exhibit compatible choices or prove the given construction commutes with every arrow.

Example 10 · A natural inclusion of singletons into power sets
Let the power-set functor use direct image. The map $s_X : X \to \mathcal{P}(X)$ sending x to $\{x\}$ is natural: direct image sends $\{x\}$ to $\{f(x)\}$, which is the same as first applying f and then s_Y. Changing the power-set arrow assignment to inverse image reverses its variance; the same displayed square is then not even typed. Naturality is always relative to specified functors, including their actions on arrows.

Failure test. A proposal to choose one distinguished element of every nonempty set naturally fails on a two-element set: its transposition would have to fix the chosen element, but fixes neither. This is a useful way to detect that objectwise choices cannot be made compatible with all symmetries.

Checkpoint 3.1. For a fixed set S, is $\eta_X : X \to X \times S$, $x \mapsto (x, s_0)$ natural in X?

Takeaways. Naturality is an equation with content. Test it on a general arrow, not just an identity.
Continue with study guide U3 and exercise sheet U3.

3.2 Lecture 6 · Functor categories and composition

Learning goals and prerequisites
Compose natural transformations, build functor categories, and interpret naturality for one-object categories.

Recall and approach. Recall composition of transformations and the distinction between an object-level arrow and a whole family of components.

For small $\mathcal{C}$ and locally small $\mathcal{D}$, functors $\mathcal{C} \to \mathcal{D}$ and natural transformations form a category $[\mathcal{C}, \mathcal{D}]$. Smallness ensures the component families form sets. Identity transformations have components 1_{FA}.

Proof strategy. Prove the component family is natural before using componentwise associativity. For horizontal composition, use the middle naturality square.

Theorem 7 · Vertical composition
If $\eta : F \Rightarrow G$ and $\theta : G \Rightarrow H$, then $(\theta\eta)_A = \theta_A \eta_A$ defines a natural transformation $F \Rightarrow H$.

Proof. For $f : A \to B$, $Hf\theta_A\eta_A = \theta_B Gf\eta_A = \theta_B\eta_B Ff$, using the two naturality equations. Associativity and identity laws hold componentwise, inherited from $\mathcal{D}$. □

One can also compose a transformation with a functor. Given $K : \mathcal{D} \to \mathcal{E}$, the components $K(\eta_A)$ form $K\eta : KF \Rightarrow KG$, because K preserves the naturality equation. Given $L : \mathcal{B} \to \mathcal{C}$, the components η_{LB} form $\eta L : FL \Rightarrow GL$. These operations are called whiskering.

Theorem 8 · Horizontal composition
For $\eta : F \Rightarrow G$ with $F, G : \mathcal{C} \to \mathcal{D}$ and $\theta : H \Rightarrow K$ with $H, K : \mathcal{D} \to \mathcal{E}$, the components $\theta_{GA} H(\eta_A)$ define a transformation $HF \Rightarrow KG$. They also equal $K(\eta_A)\theta_{FA}$.

Proof. The equality of the two expressions is naturality of θ applied to η_A. For $f : A \to B$, use naturality of θ at Gf, functoriality of H, and naturality of η to obtain $KGf\theta_{GA}H\eta_A = \theta_{GB}H\eta_B HFf$. This is the required square. □

Example 11 · Representations as functors
Regard a group G as a one-object category. A functor into vector spaces selects a vector space V and invertible linear maps $\rho(g)$ with $\rho(gh) = \rho(g)\rho(h)$. This is a representation. A natural transformation between two representations is a linear map T satisfying $T\rho(g) = \sigma(g)T$ for every g: an intertwiner.

This example shows that transformations need not be isomorphisms. The zero intertwiner is natural; an arbitrary linear map usually is not. Naturality encodes compatibility with an entire algebraic structure at once.

Example 12 · Intertwiners as a system of equations
Let a generator of C_2 act on K^2 by $D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, over a field of characteristic not two. A natural endomorphism is a matrix $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfying $TD = DT$. Comparing entries gives $b = -b$ and $c = -c$, hence $b = c = 0$. Thus exactly the diagonal matrices intertwine this action. In characteristic two, D becomes the identity and every matrix intertwines, showing that the coefficient-field hypothesis has content.

Composition of two such diagonal intertwiners is again diagonal, illustrating vertical composition concretely. Whiskering by the underlying-set functor retains the same compatible functions but forgets their linear structure. It does not assert that every equivariant set function is linear.

Checkpoint 3.2. For a group as a one-object category, what is a natural transformation between two group homomorphisms F, G into a group H?

3.2.1 A slower route through the central ideas

Example 13 · Singletons are natural; complements are not
For the covariant power-set functor, define $s_{X(x)} = \{x\}$. For a function $f : X \to Y$, the two routes send x to $f(\{x\}) = \{f(x)\}$ and to $s_{Y(f(x))} = \{f(x)\}$, so they agree. By contrast, complementation of subsets does not commute with direct image. Take the inclusion of a singleton $X = \{a\}$ into $Y = \{a, b\}$ and the empty subset of X. Complement then direct image gives $\{a\}$, while direct image then complement gives $\{a, b\}$. A formula using only sets is not automatically natural.

$$\begin{array}{ccc} X & \to s_X & P(X) \\ \downarrow f & & \downarrow P(f) \\ Y & \to s_Y & P(Y) \end{array}$$

Expanded calculation: dual evaluation. For a vector space V, define $j_V : V \to V^{**}$ by $j_{V(v)}(\varphi) = \varphi(v)$. If $f : V \to W$, its dual sends $\psi : W \to K$ to $\psi \circ f$, and its double dual has the original direction. To check $f^{**} j_V = j_W f$, evaluate both sides first at v, then at an arbitrary functional ψ on W:

$$\left(f^{**} j_{V(v)}\right)(\psi) = j_{V(v)}(\psi \circ f) = \psi(f(v)) = j_{W(f(v))}(\psi). \tag{1}$$

Equality after every functional input proves equality of the double-dual elements. This naturality calculation does not require finite dimension; surjectivity of j_V is a different claim that does.

Example 14 · Two naturality tests determine a whole vector-space family
Suppose linear maps $\eta_V : V \to V$ commute with every linear map. On K, linearity gives $\eta_{K(a)} = ac$ where $c = \eta_{K(1)}$. For $v \in V$, the map $i_{v(a)} = av$ then gives $\eta_{V(v)} = i_{v\left(\eta_{K(1)}\right)} = cv$. On K^2, the two inclusions of K therefore force both basis vectors to use this same scalar; there is no freedom to select a different scalar for each coordinate or each dimension.

A failed shortcut. Checking a square only on isomorphisms establishes invariance under changes of coordinates, which is weaker than naturality under all maps. Inclusions, projections, and maps from small test objects often carry the decisive information.

Takeaways. Functor categories let us treat whole diagrams as objects. Naturality supplies their morphisms, and componentwise reasoning proves their category laws.
Continue with study guide U3 and exercise sheet U3.

4 Unit 4 · Universal properties

4.1 Lecture 7 · Terminal objects and products

Learning goals and prerequisites
State a universal property with existence and uniqueness, prove uniqueness up to unique compatible isomorphism, and compute products in different categories.

Recall and approach. Recall pairing of functions, coordinate projections, and greatest lower bounds in an ordered set.

Definition 9 · Terminal and initial
An object T is terminal if every object X has exactly one arrow $X \to T$. An object I is initial if every X has exactly one arrow $I \to X$. A zero object is both.

A singleton is terminal in sets, while the empty set is initial. The trivial group is both initial and terminal in groups. The integers are initial among commutative unital rings; the zero ring is terminal under our convention allowing it.

Proof strategy. Build comparison maps in both directions and use uniqueness of mediators to identify their composites with identities.

Theorem 9 · Uniqueness of terminal objects
Any two terminal objects have a unique isomorphism between them.

Proof. There are unique arrows $u : T \to T'$ and $v : T' \to T$. The composite vu and identity are both arrows $T \to T$, so uniqueness forces equality. Similarly $uv = 1_{T'}$. The arrow u was already unique. □

Definition 10 · Product
A product of A, B is an object P with projections $p : P \to A$, $q : P \to B$ such that for every $f : X \to A, g : X \to B$ there is a unique $\langle f, g\rangle : X \to P$ with $p\langle f, g\rangle = f$ and $q\langle f, g\rangle = g$.

In sets the product is pairs with coordinate projections. In groups it is the Cartesian product with coordinatewise operation; the same pairing map is a homomorphism because its coordinates are. In a poset, a product is a greatest lower bound: the projections say it is below both objects, while the universal arrow says every other lower bound is below it.

Theorem 10 · Products are unique with their projections
Any two products of A, B admit a unique isomorphism preserving both projections.

Proof. Each product's universal property gives an arrow to the other preserving the projections. Each composite preserves the original projections, as does the identity. Uniqueness in the universal property forces both composites to be identities. □

Example 15 · Why the projections matter
A two-element set has several bijections to itself, so an abstract statement "products are uniquely isomorphic" is incomplete. The uniqueness refers to the isomorphism that commutes with the specified projection maps. Universal objects come with structure.

Example 16 · Deriving an equation without elements
Let p, q be product projections and let $f : X \to A$, $g : X \to B$, $h : Y \to X$. Then $\langle f, g\rangle h = \langle fh, gh\rangle$. To prove it, postcompose each side with p and obtain fh; postcompose with q and obtain gh. Product uniqueness gives equality. This proof works in every category with the product, even where the objects have no underlying elements.

The same method proves $\langle p, q\rangle = 1$ on a product. These two equations are the practical calculation rules of the universal property. When a diagram chase stalls, ask whether the desired arrows land in a product and can therefore be tested through its projections.

Checkpoint 4.1. What is the product of twelve and eighteen in the poset of positive integers ordered by divisibility?

Takeaways. A universal property is an interface: specify the structural arrows, the competing data, and the unique mediating arrow.
Continue with study guide U4 and exercise sheet U4.

4.2 Lecture 8 · Coproducts and free constructions

Learning goals and prerequisites
Dualize products, compute coproducts without confusing them with products, and use free objects through their extension property.

Recall and approach. Recall finite linear combinations and the difference between a tagged union and an overlapping ordinary union.

Definition 11 · Coproduct
A coproduct $A \coprod B$ has injections $i : A \to A \coprod B$, $j : B \to A \coprod B$. For every $f : A \to X, g : B \to X$ there is a unique $[f, g] : A \coprod B \to X$ with $[f, g]i = f$ and $[f, g]j = g$.

In sets, use a tagged disjoint union: even when A and B overlap, the tags keep their copies distinct. In vector spaces, a finite direct sum is both product and coproduct. In groups, the direct product is generally not a coproduct, because images from its two factors commute, whereas arbitrary homomorphisms from the factors into a target need not have commuting images.

Example 17 · Free vector spaces
For a set S, let K^S consist of finitely supported coefficient functions $S \to K$. The basis insertion sends s to $[s]$. Every function $u : S \to U(V)$ extends uniquely to a linear map $\overline{u} : K^S \to V$, by $\sum_s a_{s[s]} \mapsto \sum_s a_s u(s)$. Finiteness of support makes the sum meaningful.

Proof strategy. Restrict a proposed map to generators, then prove that extending those values recovers it uniquely.

Theorem 11 · Free construction determines maps
The displayed extension establishes a bijection between linear maps $K^S \to V$ and set maps $S \to U(V)$. This bijection is natural in both variables.

Proof. Restriction to basis elements and linear extension are inverse operations: a linear map is determined by its basis values, and extending then restricting returns the original function. If $a : S' \to S$, replacing each basis element by its image commutes with extension. If $b : V \to W$ is linear, applying b to a finite linear combination commutes with the sum. These are the two naturality equations. □

This is the first example of an adjunction. Notice that the defining universal property names the input set and the basis insertion; merely saying that a vector space has a basis of a given size does not specify the extension map.

Theorem 12 · Uniqueness of a free object
Two free vector spaces on the same set, with their specified basis insertions, are uniquely isomorphic compatibly with those insertions.

Proof. Extend each insertion into the other free object. Their composites agree with the respective identity on all generators, so uniqueness of extension makes the composites identities. □

Example 18 · Free groups and why new words appear
A free group on a set S has an insertion $S \to U(FS)$ such that every function $S \to U(G)$ extends uniquely to a homomorphism $FS \to G$. One construction uses words in letters s, s^{-1}, modulo insertion and deletion of adjacent inverse pairs. Concatenation is associative before quotienting and respects this equivalence, the empty word is an identity, and reversing a word and inverting its letters supplies an inverse. Evaluating a word in G respects the cancellations, giving the unique extension. This is the free-group property used later.

For two free generators, a map is determined by two arbitrary target elements, even when they do not commute. Their product and longer alternating words must therefore be represented in the free group. A direct product would force commuting images and cannot implement that mapping problem. This compares two universal properties through the maps they permit.

Checkpoint 4.2. Why does the direct sum of infinitely many vector spaces require finite support?

Takeaways. Universal properties organize existence and uniqueness of maps. Concrete constructions verify that such objects actually exist in the category at hand.
Continue with study guide U4 and exercise sheet U4.

5 Unit 5 · Limits and colimits

Bridge into this unit. Products collect independent coordinates; equalizers impose compatibility between those coordinates. A general limit does both. The dual construction starts with independently chosen summands and then identifies the pairs demanded by the diagram.

5.1 Lecture 9 · Cones, equalizers, and pullbacks

Learning goals and prerequisites
State the limit of a diagram, compute elementary limits, and translate cone equations into elementwise constructions in sets.

Recall and approach. Recall products, subsets defined by equations, and uniqueness through projections.

Definition 12 · Diagram, cone, limit
A diagram of shape J in $\mathcal{C}$ is a functor $D : J \to \mathcal{C}$. A cone with apex X has maps $\lambda_j : X \to Dj$ satisfying $D(a)\lambda_i = \lambda_j$ for every $a : i \to j$. A limit is a cone $\left(L, p_j\right)$ such that every cone factors uniquely through it: $\lambda_j = p_j u$ for a unique $u : X \to L$.

The empty diagram has a limit precisely when the category has a terminal object. A discrete two-object diagram has a product as its limit. Thus products are a special instance of a single general pattern.

Definition 13 · Equalizer and pullback
An equalizer of $f, g : A \to B$ is $e : E \to A$ with $fe = ge$, universal among arrows with that equality. A pullback of $f : A \to C$ and $g : B \to C$ is a square $P \to B$, $P \to A$ with equal composites to C, universal among such squares.

In sets, the equalizer is the subset $\{a : f(a) = g(a)\}$ with inclusion. Given an arrow $u : X \to A$ equalizing the maps, every value lies in this subset, so restricting its codomain supplies the unique factor. For a pullback use $P = \{(a, b) : f(a) = g(b)\}$ with coordinate projections. The unique mediator of compatible maps is $x \mapsto (u(x), v(x))$.

Proof strategy. Translate each compatibility equation into a condition on coordinates, then prove every competing cone factors uniquely.

Theorem 13 · Equalizers are monic
Every equalizer arrow is a monomorphism.

Proof. If $eu = ev$, then $u, v : X \to E$ are both mediators for the same equalizing arrow $eu : X \to A$. Uniqueness in the equalizer property implies $u = v$. □

Example 19 · Fibers as pullbacks
A chosen element $c \in C$ is a map $1 \to C$. Pulling $f : A \to C$ back along this map gives the fiber $f^{-1}(c)$. Pullbacks therefore generalize inverse-image constructions without referring to elements.

The slogan "a limit is a compatible tuple" is useful in sets: it is the subset of $\prod_j Dj$ satisfying all arrow equations. In groups and vector spaces these equations define a subgroup or subspace, and the same universal property follows because the coordinate maps preserve operations.

Example 20 · A pullback can be empty
Let $A = \{a\}$ and $B = \{b\}$ map into $C = \{0, 1\}$ by $a \mapsto 0$ and $b \mapsto 1$. No pair has matching images, so their pullback in sets is empty. A cone from a nonempty X cannot exist: at any x the compatibility equation would assert $0 = 1$. A cone from the empty set has a unique mediator to the empty pullback. Thus an empty candidate can satisfy a universal property; existence of arrows in the competing data must be checked before demanding a nonempty answer.

Distinction. An equalizer is a map into the common source, while a pullback comes with two maps into two sources. Both impose equations, but the diagram shapes and competing data differ.

Checkpoint 5.1. What is the equalizer of two distinct parallel arrows in a poset category?

Takeaways. Specify the shape of the diagram and every compatibility equation before proposing its limit.
Continue with study guide U5 and exercise sheet U5.

5.2 Lecture 10 · Building limits and understanding colimits

Learning goals and prerequisites
Construct limits from products and equalizers, dualize to colimits, and recognize quotient constructions as coequalizers.

Recall and approach. Recall small indexing categories and the equivalence relation generated by a set of pairs.

Theorem 14 · Products and equalizers construct limits
A category with all small products and equalizers has all small limits. Finite products and equalizers suffice for finite limits.

Proof strategy. Put all candidate coordinates in one product and impose all compatibility conditions with one equalizer. **Proof.** For $D : J \to \mathcal{C}$, form $P = \prod_{j \in J} Dj$ and $Q = \prod_{a : i \to j} Dj$. Define $s, t : P \to Q$ by the components $s_a = D(a)\pi_i$ and $t_a = \pi_j$. Let $e : L \to P$ equalize s, t. Its composites with the product projections form a cone. Any cone defines a unique map $X \to P$, and its cone equations say this map equalizes s, t, so it factors uniquely through L. Uniqueness follows successively from equalizer and product uniqueness. If the indexing category has finitely many objects and arrows, the products needed are finite. □

Definition 14 · Colimit
A cocone consists of maps $i_j : Dj \to C$ with $i_j D(a) = i_i$ for $a : i \to j$. A colimit is a cocone through which every other cocone factors uniquely by an arrow from its apex. It is a limit in the opposite category.

In sets, a colimit of a small diagram is obtained from the disjoint union of all its sets by identifying each element with its image under every diagram arrow, then taking the equivalence relation generated by those identifications. A cocone is exactly a function on the disjoint union that is constant on these equivalence classes, which proves the quotient's universal property.

Example 21 · Coequalizers and pushouts
The coequalizer of $f, g : A \to B$ in sets is the quotient of B by the equivalence relation generated by $f(a) \sim g(a)$. A pushout of $A \leftarrow C \to B$ is the tagged disjoint union of A, B with the images of each c identified. Taking only the listed pairs without reflexive, symmetric, and transitive closure generally fails.

Theorem 15 · Coproducts and coequalizers construct colimits
A category with all small coproducts and coequalizers has all small colimits.

Proof. Apply the preceding limit theorem to the opposite category. Explicitly, the two arrows from the coproduct over diagram arrows to the coproduct over objects encode respectively the source element and its image. Their coequalizer enforces exactly the cocone equations. □

Example 22 · Computing a pushout of vector spaces
For $f : U \to V$ and $g : U \to W$, form $(V \oplus W)/N$, where N is the image of $u \mapsto (f(u), -g(u))$. The two summand maps become equal on the images of U. Given linear maps $a : V \to X$, $b : W \to X$ with $af = bg$, the sum map $(v, w) \mapsto a(v) + b(w)$ kills N and therefore factors through the quotient. Every quotient element has a representative pair, so the factor is unique.

This calculation is the linear counterpart of gluing tagged sets, but it uses a subspace of relations rather than just an equivalence relation on points. Substituting the set construction directly into vector spaces would miss the requirement that all linear consequences of the relations vanish.

Checkpoint 5.2. In vector spaces, what is the coequalizer of $f, g : V \to W$?

5.2.1 A slower route through the central ideas

Example 23 · A pullback as an intersection
Let A, B be subsets of X, with inclusion maps to X. Their pullback consists of pairs (a, b) with $a = b$ in X, and hence is naturally identified with $A \cap B$. If maps $u : T \to A$ and $v : T \to B$ have equal composites into X, their common value lies in the intersection and defines the unique mediator. Notice that the intersection description depends on the maps being inclusions; arbitrary maps can have distinct elements with the same image.

$$\begin{array}{ccc} A \cap B & \to \text{inclusion} & B \\ \downarrow \text{inclusion} & & \downarrow \text{inclusion} \\ A & \to \text{inclusion} & X \end{array}$$

Expanded proof: building a limit from equations. For a small diagram $D : J \to \mathcal{C}$, take $P = \prod_{j \in J} D(j)$ and $Q = \prod_{u : i \to j} D(j)$, assuming these products exist. Define two maps from P to Q: in coordinate $u : i \to j$, the first is $D(u)\pi_i$, the second is π_j. Their equalizer $e : L \to P$ has legs $\pi_j e$, and the equalizer equation is precisely the cone equation for every arrow u. Any cone $\big(t_j : T \to D(j)\big)$ first gives a unique $t : T \to P$ by the product property. Its cone equations say the two maps agree on t, so the equalizer gives a unique lift $T \to L$. Product uniqueness and equalizer uniqueness together prove the limit property.

If J has no objects, the first product is terminal, so the construction includes the empty limit. The products are indexed by sets because the diagram is small; this is where the size assumption is used.

Example 24 · Why a coequalizer needs transitive identifications
Take maps from $\{u, v\}$ to $\{a, b, c\}$ with $f(u) = a, g(u) = b, f(v) = b, g(v) = c$. A coequalizing function must assign equal values to a, b and to b, c, hence to all three. The coequalizer is the singleton quotient. Merely listing the two prescribed pairs is not an equivalence relation: transitivity supplies the missing identification of a with c. A map constant on the resulting class factors uniquely through the singleton.

A failed shortcut. A set of compatible tuples provides a construction in sets, but does not automatically construct limits in every category. The product-and-equalizer proof identifies the categorical ingredients that must actually exist.

Takeaways. Limits impose equations inside products; colimits impose identifications on coproducts. The statements are dual even when the concrete constructions look very different.
Continue with study guide U5 and exercise sheet U5.

6 Unit 6 · Representability and Yoneda

Bridge into this unit. Naturality looks like infinitely many commuting squares. Yoneda shows that, for a representable source, one square involving each arrow out of a fixed object determines all values from the identity arrow. Learn the reconstruction before treating the lemma as a counting device.

6.1 Lecture 11 · The Yoneda correspondence

Learning goals and prerequisites
State and prove the covariant Yoneda lemma, calculate its two inverse maps, and interpret naturality as the reason one element determines a transformation.

Recall and approach. Recall hom functors and the naturality square for a map whose source is the representing object.

Let $\mathcal{C}$ be locally small and let $F : \mathcal{C} \to \mathrm{Set}$. For fixed A, a transformation from $\mathcal{C}(A, -)$ to F appears to contain one function for every object. Yoneda says this whole family is determined by one element of FA.

Theorem 16 · Covariant Yoneda lemma
There is a bijection $\mathrm{Nat}(\mathcal{C}(A, -), F) \cong FA$ sending η to $\eta_{A(1_A)}$. Its inverse sends $x \in FA$ to the transformation $\eta^x_{B(f)} = F(f)(x)$.

Proof strategy. Use the identity arrow as the universal test case. Naturality at an arbitrary $f : A \to B$ then forces every other value. **Proof.** For $g : B \to C$, $F(g)\eta^x_{B(f)} = F(g)F(f)(x) = F(gf)(x) = \eta^x_{C(gf)}$, so the proposed family is natural. Evaluating it at 1_A returns $F(1_A)(x) = x$. Conversely, naturality of any η at $f : A \to B$ gives $F(f)\eta_{A(1_A)} = \eta_{B(f1_A)} = \eta_{B(f)}$. Thus rebuilding from $\eta_{A(1_A)}$ returns the entire original transformation. The two operations are inverse. □

The bijection is natural in F: composing η with $\theta : F \Rightarrow G$ sends its distinguished element to $\theta_{A\left(\eta_{A(1_A)}\right)}$. It is also natural in A with the appropriate variance: a map $a : A \to A'$ induces precomposition $\mathcal{C}(A', -) \to \mathcal{C}(A, -)$, and the corresponding element changes by $F(a)$.

Example 25 · Natural operations on pairs
In sets, $\mathrm{Set}(\{0, 1\}, X)$ identifies with $X \times X$. Natural transformations from this functor to the identity functor correspond to elements of $\{0, 1\}$. There are exactly two: the two coordinate projections. A rule that "chooses the larger coordinate" is not natural for arbitrary set maps and is not even defined on unstructured sets.

Watch for this. The identity used in Yoneda is the identity of the representing object, not an identity element of FA. The functor's values are sets and need not have algebraic operations.

Example 26 · Classifying a natural binary operation with a larger target
Consider a natural family $X \times X \to \mathcal{P}(X)$, where the power-set functor uses direct image. Since the pair functor is represented by the two-element set 2, Yoneda identifies these transformations with the four subsets of 2. They give the operations $(x, y) \mapsto \varnothing$, $\{x\}$, $\{y\}$, and $\{x, y\}$. Reconstruction sends a selected subset of the index set along the function $2 \to X$ naming the input pair.

This classification includes repeated inputs: if $x = y$, the image of the two-element subset is a singleton, exactly as direct image requires. Naturality determines behavior under identifications as well as under bijections. Testing only permutations would miss that constraint.

Checkpoint 6.1. What element corresponds to the identity transformation of $\mathcal{C}(A, -)$?

Takeaways. To classify natural transformations out of a representable functor, evaluate at the identity and then reconstruct by functoriality.
Continue with study guide U6 and exercise sheet U6.

6.2 Lecture 12 · Representing objects and universal elements

Learning goals and prerequisites
Use Yoneda to prove uniqueness of representations, distinguish covariant and contravariant versions, and recognize universal constructions as representability.

Recall and approach. Recall the covariant lemma and that replacing a category by its opposite reverses the relevant hom arguments.

Definition 15 · Representable functor
A covariant functor $F : \mathcal{C} \to \mathrm{Set}$ is representable if it is naturally isomorphic to $\mathcal{C}(A, -)$ for some A. The corresponding element $u \in FA$ is universal: for every $x \in FX$, there is exactly one $f : A \to X$ with $F(f)(u) = x$.

The equivalence of the two descriptions follows directly from the inverse map in Yoneda: the component at X sends f to $F(f)(u)$ and is bijective precisely under the stated existence-and-uniqueness condition.

For the presheaf-category statement below, take $\mathcal{C}$ small (or work in a larger universe). The preceding Yoneda correspondence itself only needs locally small hom sets.

Proof strategy. Translate the natural transformations to representing arrows with Yoneda, then use faithfulness on the inverse equations.

Theorem 17 · Yoneda embedding
For $h_A = \mathcal{C}(-, A)$, arrows $A \to B$ correspond bijectively to natural transformations $h_A \Rightarrow h_B$, by postcomposition. Thus $A \mapsto h_A$ defines a full and faithful functor into the presheaf category $[\mathcal{C}^{\mathrm{op}}, \mathrm{Set}]$.

Proof. Apply the contravariant version of Yoneda, obtained from the covariant lemma in $\mathcal{C}^{\mathrm{op}}$: a transformation is determined by its value at 1_A, an element of $\mathcal{C}(A, B)$. Reconstruction sends $u : X \to A$ to the composite fu. Identity and composition of transformations correspond to those of arrows. □

Theorem 18 · Uniqueness of representing objects
If h_A and h_B are naturally isomorphic, then A and B are isomorphic. The analogous result holds for covariant representables.

Proof. Fullness supplies arrows corresponding to the natural isomorphism and its inverse. Faithfulness turns the two identity equations for their composites into the corresponding identity equations for the arrows. In the covariant case the arrow directions reverse, with the same conclusion. □

Example 27 · Products as representability
For fixed A, B, the functor $X \mapsto \mathcal{C}(X, A) \times \mathcal{C}(X, B)$ is contravariant. A product P represents it through the natural bijection $\mathcal{C}(X, P) \cong \mathcal{C}(X, A) \times \mathcal{C}(X, B)$. The universal pair of projections is the image of 1_P.

For a covariant F, its category of elements has objects (A, x) with $x \in FA$ and arrows $(A, x) \to (B, y)$ given by $f : A \to B$ satisfying $F(f)(x) = y$. A universal element is exactly an initial object of this category. This reformulates representability as a familiar universal-property question.

Example 28 · A universal element carries structural maps
For the contravariant functor $X \mapsto \mathcal{C}(X, A) \times \mathcal{C}(X, B)$, an element at P is a pair of arrows $p : P \to A, q : P \to B$. Its universal property says every pair from X is uniquely obtained by precomposing these two arrows with $u : X \to P$. This is exactly the product property, including the projections. The representing object alone is insufficient data; the universal element supplies its interface.

For a covariant representable $\mathcal{C}(A, -)$, a transformation to $\mathcal{C}(B, -)$ is determined by an element of $\mathcal{C}(B, A)$, hence by an arrow $B \to A$. For contravariant representables the corresponding arrow goes $A \to B$. Writing this one identity-component calculation prevents a common variance error.

Checkpoint 6.2. Why is the underlying-set functor from groups to sets represented covariantly by $\mathbb{Z}$?

6.2.1 A slower route through the central ideas

Expanded proof: reconstructing a transformation. Fix $F : \mathcal{C} \to \mathrm{Set}$ and A. Given $x \in F(A)$, put $\eta^x_{B(f)} = F(f)(x)$. For $g : B \to C$, functoriality gives

$$F(g)\big(\eta^x_{B(f)}\big) = F(g)(F(f)(x)) = F(g \circ f)(x) = \eta^x_{C(g\circ f)}. \tag{2}$$

Thus the definition is natural. Evaluation at the identity recovers x because $F(1_A) = \mathrm{id}$. In the other direction, the naturality square of any η at $f : A \to B$, evaluated at 1_A, gives

$$F(f)\big(\eta_{A(1_A)}\big) = \eta_{B(f\circ 1_A)} = \eta_{B(f)}. \tag{3}$$

This is the exact line explaining why no other values can be chosen independently. Both inverse identities have now been proved.

$$\begin{array}{ccc} \mathcal{C}(A,A) & \to \eta_A & F(A) \\ \downarrow \mathcal{C}(A,f) & & \downarrow F(f) \\ \mathcal{C}(A,B) & \to \eta_B & F(B) \end{array}$$

Example 29 · The generic ordered pair
The functor $X \mapsto X \times X$ is represented by the two-element set $D = \{0,1\}$: a function $D \to X$ is exactly its pair of values. To classify natural functions $X \times X \to X$, use the preceding reconstruction with $F = \mathrm{id}$. The generic pair $(0,1)$ must be sent to either zero or one. Naturality then forces the first projection or the second projection on every set. There is no natural average on arbitrary sets because no such extra choice exists at the representing object.

Example 30 · A group element as a map out of the integers
For the forgetful functor $U : \mathrm{Grp} \to \mathrm{Set}$, a group homomorphism $\mathbb{Z} \to G$ is uniquely determined by the image of one, with $n \mapsto g^n$. Hence U is represented by $\mathbb{Z}$. A natural endomorphism of U is determined by an integer k, and on each group its formula is $g \mapsto g^k$. These are functions, not necessarily group homomorphisms. For example squaring is natural with respect to group homomorphisms but is not itself a homomorphism on every nonabelian group.

A failed shortcut. Confusing the components' category with the input category changes the problem. The group example has components in sets, so naturality does not assert $(gh)^k = g^k h^k$. Always type the component before imposing algebraic conditions on it.

Takeaways. Representability turns an element-valued question into a mapping question. Yoneda explains why the resulting representing object is determined up to compatible isomorphism.
Continue with study guide U6 and exercise sheet U6.

7 Unit 7 · Adjunctions

Bridge into this unit. An adjunction compares two ways of specifying a map, usually a free construction and a structured target. Its unit and counit arise by translating identity maps. The triangle identities say that translating out and back does not change the original data.

7.1 Lecture 13 · Natural hom-set bijections

Learning goals and prerequisites
Recognize an adjunction, compute adjunct transposes, and verify naturality in both variables.

Recall and approach. Recall universal properties of free objects and the fact that a poset hom set is either empty or a singleton.

An adjunction formalizes a reversible change in the form of a problem. A map out of a free object is equivalent to an unconstrained map on generators; a map from a product is equivalent to a map into a function object. The objects on the two sides need not be isomorphic, but their mapping problems correspond naturally.

Definition 16 · Adjunction
Functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ satisfy $F \dashv G$ if there are bijections $\Phi_{C,D} : \mathcal{D}(FC, D) \cong \mathcal{C}(C, GD)$ natural in C, D. The image of a map under this bijection is its adjunct transpose.

Naturality means two equations. For $a : C' \to C$, $\Phi(hFa) = \Phi(h)a$. For $b : D \to D'$, $\Phi(bh) = G(b)\Phi(h)$. These equations make it possible to move calculations through the bijection rather than treating each hom set independently.

Example 31 · Free and forgetful
Let F send a set to its free vector space and U forget vector-space structure. Then $\mathrm{Vect}_{K(FX,V)} \cong \mathrm{Set}(X, UV)$ by restriction to basis vectors and linear extension. The two naturality equations are respectively substitution of generators and composition with a linear map. Thus $F \dashv U$.

Example 32 · Product and exponential in sets
For fixed A, let X^A denote functions $A \to X$. There is a bijection $\mathrm{Set}(B \times A, X) \cong \mathrm{Set}\big(B, X^A\big)$ sending f to $b \mapsto (a \mapsto f(b, a))$. Its inverse evaluates: $g \mapsto ((b, a) \mapsto g(b)(a))$. Substitution verifies naturality, so $(- \times A) \dashv (-)^A$.

Proof strategy. Give restriction/extension or curry/evaluate maps explicitly, check both composites, then check each variable of naturality.

Theorem 19 · Adjoints in posets
For monotone maps $F : P \to Q$ and $G : Q \to P$, an adjunction is exactly the condition $F(p) \leq q$ if and only if $p \leq G(q)$.

Proof. Each hom set in a poset is empty or a singleton. A bijection between the two hom sets exists exactly when they are simultaneously empty or simultaneously inhabited, giving the stated equivalence. Naturality is automatic because there is at most one parallel arrow. □

For a function $f : X \to Y$, direct image and inverse image give such an adjunction between power-set posets: $f[S] \subseteq T$ exactly when $S \subseteq f^{-1}[T]$. This elementary logical equivalence is the same formal pattern as free vector spaces and currying.

Example 33 · Two adjoints around inverse image
For a function $f : X \to Y$, inverse image f^* on power sets also has a right adjoint: send $S \subseteq X$ to the set of y whose entire fiber $f^{-1}(y)$ lies in S. Indeed $f^*(T) \subseteq S$ means every element over any $y \in T$ lies in S, exactly that T lies in the stated set. Empty fibers satisfy this condition automatically.

Thus direct image is an existential operation on fibers and this right adjoint is a universal operation on fibers. Both comparisons arise from explicit inclusion equivalences. The difference at empty fibers is useful: existential image excludes a point with no witness, while the universal condition holds there vacuously.

Checkpoint 7.1. What is the adjunct transpose of $f : B \times A \to X$ evaluated at b?

Takeaways. To prove an adjunction, give both directions of the hom-set correspondence, prove they are inverse, and verify naturality.
Continue with study guide U7 and exercise sheet U7.

7.2 Lecture 14 · Units, counits, and triangle identities

Learning goals and prerequisites
Recover the unit and counit from a hom-set adjunction and prove the converse construction.

Recall and approach. Recall the two-variable naturality equations of an adjunction and how identity arrows determine unit and counit.

Definition 17 · Unit and counit
For $F \dashv G$, set $\eta_C = \Phi_{C,FC}(1_{FC}) : C \to GFC$ and $\varepsilon_D = \Phi^{-1}_{GD,D}(1_{GD}) : FGD \to D$.

The unit inserts an object into the result of going across and back; the counit evaluates the result of the opposite round trip. They are transformations $\eta : 1_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1_{\mathcal{D}}$.

Proof strategy. Use naturality to move a map past the unit or counit, then apply the matching triangle identity.

Theorem 20 · Transpose formulas and triangles
For $h : FC \to D$ and $k : C \to GD$,

$$\Phi(h) = G(h)\eta_C, \quad \Phi^{-1}(k) = \varepsilon_D F(k). \tag{4}$$

Moreover $\varepsilon_{FC} F(\eta_C) = 1_{FC}$ and $G(\varepsilon_D)\eta_{GD} = 1_{GD}$.

Proof. The first formula is naturality in D applied to the identity of FC. The second is naturality in C applied to the identity of GD. Applying the inverse correspondence to η_C gives the first triangle, and applying the correspondence to ε_D gives the second. Naturality of η follows by evaluating both paths on an arrow $a : C \to C'$ and using the two naturality equations for Φ; the argument for ε is dual. □

Theorem 21 · Converse
If natural transformations $\eta : 1 \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1$ satisfy the two triangle identities, the displayed transpose formulas define an adjunction.

Proof. Starting with $h : FC \to D$, going across and back gives $\varepsilon_D FG(h)F(\eta_C)$. Naturality of ε replaces its first two factors by $h\varepsilon_{FC}$; the first triangle then gives h. Starting with $k : C \to GD$, the round trip is $G(\varepsilon_D)GF(k)\eta_C$. Naturality of η makes this $G(\varepsilon_D)\eta_{GD}k = k$ by the second triangle. The formulas are natural by functoriality and naturality of the two transformations. □

Example 34 · Free vector spaces again
The unit maps a set element to its basis vector. The counit sends a formal linear combination of underlying vectors to their actual sum in the vector space. The first triangle says that expanding a basis vector and then evaluating does nothing; the second says that inserting one actual vector and evaluating returns it.

Watch for this. The unit and counit need not be isomorphisms. An adjunction is generally weaker than an equivalence. Their triangles express coherence, not invertibility of the round-trip functors.

Example 35 · Both triangles for product and exponential
Fix A in sets. The unit at X sends x to the function $a \mapsto (x, a)$ in $(X \times A)^A$. The counit at Y evaluates (f, a) to $f(a)$. The first triangle starts with (x, a), turns x into that function, retains a, and evaluates, returning (x, a). The second starts with $f : A \to Y$, builds the function $a \mapsto (f, a)$, and applies evaluation pointwise, returning $a \mapsto f(a)$, the original set function.

These computations explain what the abstract triangles control. They do not say the unit is onto: many functions $A \to X \times A$ are not of the special form $a \mapsto (x, a)$. An adjunction can have coherent round trips without making every individual unit or counit invertible.

Checkpoint 7.2. For product–exponential adjunction, what is the counit at X?

7.2.1 A slower route through the central ideas

Example 36 · Words and the free-group adjunction
Let F be the free-group functor and U forget the group structure. A set map $a : X \to U(G)$ extends to the homomorphism $|a| : F(X) \to G$ sending a reduced word $x_1^{\varepsilon_1} \dots x_r^{\varepsilon_r}$ to $a(x_1)^{\varepsilon_1} \dots a(x_r)^{\varepsilon_r}$. Adjacent inverse letters evaluate to inverse elements, so reductions do not change the result. Conversely restrict a homomorphism to the one-letter generators. Restriction and extension are inverse because homomorphisms preserve products and inverses, and every word is built from generators.

Expanded derivation of the triangles. The unit sends x to its one-letter word in $UF(X)$. The counit $\varepsilon_G : FU(G) \to G$ evaluates a word whose letters are elements of G. For the first triangle, a generator x of $F(X)$ becomes the generator named by its one-letter word, then evaluation returns the original generator. Equality on generators proves equality of homomorphisms. For the second, an element g of $U(G)$ becomes its one-letter word and is evaluated back to g. This explains the two triangles by following data rather than memorizing their shapes.

$$\begin{array}{ccc} F(X) & \to F(\eta_X) & FUF(X) \\ \downarrow \mathrm{id} & & \downarrow \varepsilon_{F(X)} \\ F(X) & \to \mathrm{id} & F(X) \end{array}$$

Example 37 · Rounding as two order adjunctions
Let $i : \mathbb{Z} \to \mathbb{R}$ be inclusion. For real r and integer n, $\lceil r \rceil \leq n$ exactly when $r \leq i(n)$, so ceiling is left adjoint to inclusion. Also $i(n) \leq r$ exactly when $n \leq \lfloor r \rfloor$, so floor is right adjoint. For $r = 2.4$, the unit inequality of the first adjunction is $2.4 \leq 3$; the counit inequality of the second is $2 \leq 2.4$. The asymmetry explains why reversing ceiling and floor would reverse a required inequality.

From a bijection to unit and counit. For an arbitrary natural bijection $\Phi_{A,B} : \mathcal{D}(FA, B) \cong \mathcal{C}(A, GB)$, define $\eta_A = \Phi_{A,FA}(1_{FA})$ and $\varepsilon_B = \Phi^{-1}_{GB,B}(1_{GB})$. Naturality forces the forward translation of h to be $G(h)\eta_A$ and the inverse translation of k to be $\varepsilon_B F(k)$. Substituting an identity into the two inverse equalities yields the triangles. Their role is to certify that these formulas really undo each other.

A failed shortcut. Finding a bijection separately for each pair of objects is insufficient. Naturality requires those bijections to commute with precomposition and postcomposition; without it, the identity-derived formulas need not reconstruct the chosen bijections.

Takeaways. The hom-set and unit–counit descriptions are interchangeable. Use whichever makes a given construction easier to verify.
Continue with study guide U7 and exercise sheet U7.

8 Unit 8 · Preservation and applications

8.1 Lecture 15 · Why adjoints preserve universal constructions

Learning goals and prerequisites
Prove that right adjoints preserve limits, use the dual theorem, and distinguish preservation from reflection.

Recall and approach. Recall that preservation concerns an existing universal cone and reflection runs the implication in the opposite direction.

Definition 18 · Preservation
A functor preserves a limit if applying it to a limiting cone gives a limiting cone of the image diagram. It reflects a limit if a cone whose image is limiting must already have been limiting. These are different implications.

Theorem 22 · Right adjoints preserve limits
If $F \dashv G$ and $D : J \to \mathcal{D}$ has limit L, then the image cone with apex GL is a limit of GD. Dually, left adjoints preserve colimits.

Proof strategy. Convert a competing cone through the adjunction, use the original universal property, then transpose the unique mediator back. **Proof.** Given a cone $u_j : X \to GDj$, transpose each arrow to $\overline{u}_j : FX \to Dj$. Naturality of the adjunction in its second variable turns the cone equations for u_j into cone equations for the transposes. Since L is a limit, there is a unique $v : FX \to L$ with $p_j v = \overline{u}_j$. Its transpose $w : X \to GL$ satisfies $G(p_j)w = u_j$, again by naturality. Any other such w would transpose to another mediator v, so it is equal. This proves the image cone is limiting. Apply this argument to opposite categories for the dual statement. □

Example 38 · Forgetting group structure
The free-group functor is left adjoint to the underlying-set functor. Therefore the underlying-set functor preserves products and equalizers. Concretely, a product group has the Cartesian product as its underlying set and an equalizer subgroup has the expected subset as its underlying set.

It does not follow that this forgetful functor preserves coproducts. The coproduct of groups is a free product, whose underlying set contains new words. It is usually much larger than the disjoint union of the underlying sets. The direction of an adjunction predicts which universal constructions are preserved.

Theorem 23 · Representables preserve limits
For fixed A, the functor $\mathcal{C}(A, -)$ preserves every existing small limit.

Proof. A map from A to a limit is equivalent to a compatible family of maps from A to the diagram objects, by the limit property. That compatible-family set is exactly the limit in sets of the hom-set diagram. The correspondence respects its projections, which is the required preservation statement. □

Example 39 · Equalizers survive the underlying-set functor
For linear maps $f, g : V \to W$, their equalizer is the kernel of $f - g$ with its inclusion. The underlying set is exactly $\{v : f(v) = g(v)\}$, the set-theoretic equalizer. A set map into this subset need not be linear, but the adjunction proof nevertheless establishes the universal property for all set cones by translating them to maps out of free vector spaces. This explains why merely testing linear competing maps would not suffice to prove preservation into sets.

Failure prediction. A free vector space on a singleton is nonzero, while the terminal vector space is zero. Therefore the free functor cannot be a right adjoint as well to the same underlying-set functor merely by symmetry of notation. The direction of the hom-set bijection matters.

Checkpoint 8.1. Does a left adjoint necessarily preserve a terminal object?

Takeaways. Preservation theorems are practical tests. Before computing a complicated image construction, ask whether an adjunction or representability already determines it.
Continue with study guide U8 and exercise sheet U8.

8.2 Lecture 16 · Connecting algebra, logic, and universal maps

Learning goals and prerequisites
Use adjunctions and representability to organize examples from algebra and logic, and synthesize the course's proof methods.

Recall and approach. Recall commutators, quotient factorization, Yoneda, and the unit–counit interpretation of function objects.

Example 40 · Abelianization as a left adjoint
For a group G, quotient by its commutator subgroup to obtain G^{ab}. A homomorphism from G into an abelian group kills every commutator and therefore factors uniquely through G^{ab}. This gives a natural bijection $\mathrm{Ab}\big(G^{\mathrm{ab}}, A\big) \cong \mathrm{Grp}(G, IA)$, where I includes abelian groups into groups. Thus abelianization is left adjoint to inclusion.

The induced map on abelianizations is well defined because homomorphisms take commutators to commutators. Naturality of the displayed bijection is not an extra miracle: composing a factorization with maps on either side produces the unique factorization of the corresponding composite.

Proof strategy. Express the same mapping problem in two ways and use naturality plus uniqueness to obtain the canonical comparison.

Theorem 24 · Free objects transport coproducts
The free vector space on a disjoint union is canonically isomorphic to the direct sum of the free vector spaces on the parts.

Proof. A linear map from $F(A \coprod B)$ to V corresponds to a function from the disjoint union to UV, hence to a pair of functions from A, B, hence to a pair of linear maps from FA, FB. These correspond uniquely to a linear map from $FA \oplus FB$. The resulting correspondence is natural in V, so covariant representability gives a compatible isomorphism. Concretely it sends each tagged basis element to the corresponding summand's basis vector. □

Definition 19 · Cartesian closed category
A category is Cartesian closed if it has finite products and, for every A, the functor $- \times A$ has a right adjoint. The resulting exponential B^A represents maps from a product: $\mathcal{C}(X \times A, B) \cong \mathcal{C}\big(X, B^A\big)$.

Sets form a Cartesian closed category. This supplies the categorical pattern behind function types: abstraction transposes a map, and application uses evaluation. The two inverse-transpose identities correspond to beta and eta laws. Type Theory will distinguish these semantic equalities from the precise syntactic conversion rules of a chosen calculus.

Example 41 · A poset version of implication
In the power-set poset of a fixed set S, products are intersections. For subsets A, B, the right adjoint to $- \cap A$ sends B to $(S \setminus A) \cup B$, because $X \cap A \subseteq B$ exactly when $X \subseteq (S \setminus A) \cup B$. This is implication expressed as an adjunction.

To solve a new categorical problem, choose among four routes. For a universal property, write the mediating map and prove uniqueness. For a natural operation, test the identity through Yoneda. For a change in mapping problem, seek an adjunction. For a preservation claim, transport competing cones through a hom-set correspondence. These methods should become reusable proof habits rather than slogans.

Example 42 · An explicit abelianization calculation
In S_3, every commutator has sign one, so the commutator subgroup lies in A_3. The commutator of (12) and (23) is a nontrivial three-cycle and generates A_3, so the commutator subgroup equals A_3. Thus abelianization gives C_2. Every homomorphism from S_3 to an abelian group is consequently determined by an element whose square is one: factor through C_2 and choose the image of its generator.

This is simultaneously a quotient computation and an adjunction application. To check the claimed universality directly, show the map kills every commutator, then use the first isomorphism theorem's factorization argument. To prove naturality, compare the two candidate factorizations of the same composite and invoke uniqueness. Neither step requires choosing representatives of all cosets by hand.

Checkpoint 8.2. Why is an isomorphism obtained from a universal property called canonical?

Takeaways. Category theory makes the dependencies among constructions explicit. Its strongest conclusions come from carefully stated universal properties and naturality equations.
Continue with study guide U8 and exercise sheet U8.